

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data columns tracked

37

Missing Values

Review ratings imputed

Customer Data

- Demographics (Age, Gender, Location)
- Subscription Status
- Previous Purchases

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Shipping Type

Behavior Metrics

- Discount Applied
- Promo Code Used
- Review Rating

Python Data Preparation

01

Data Loading

Imported dataset using pandas

02

Exploration

Checked structure and summary statistics

03

Cleaning

Handled missing values, standardized columns

04

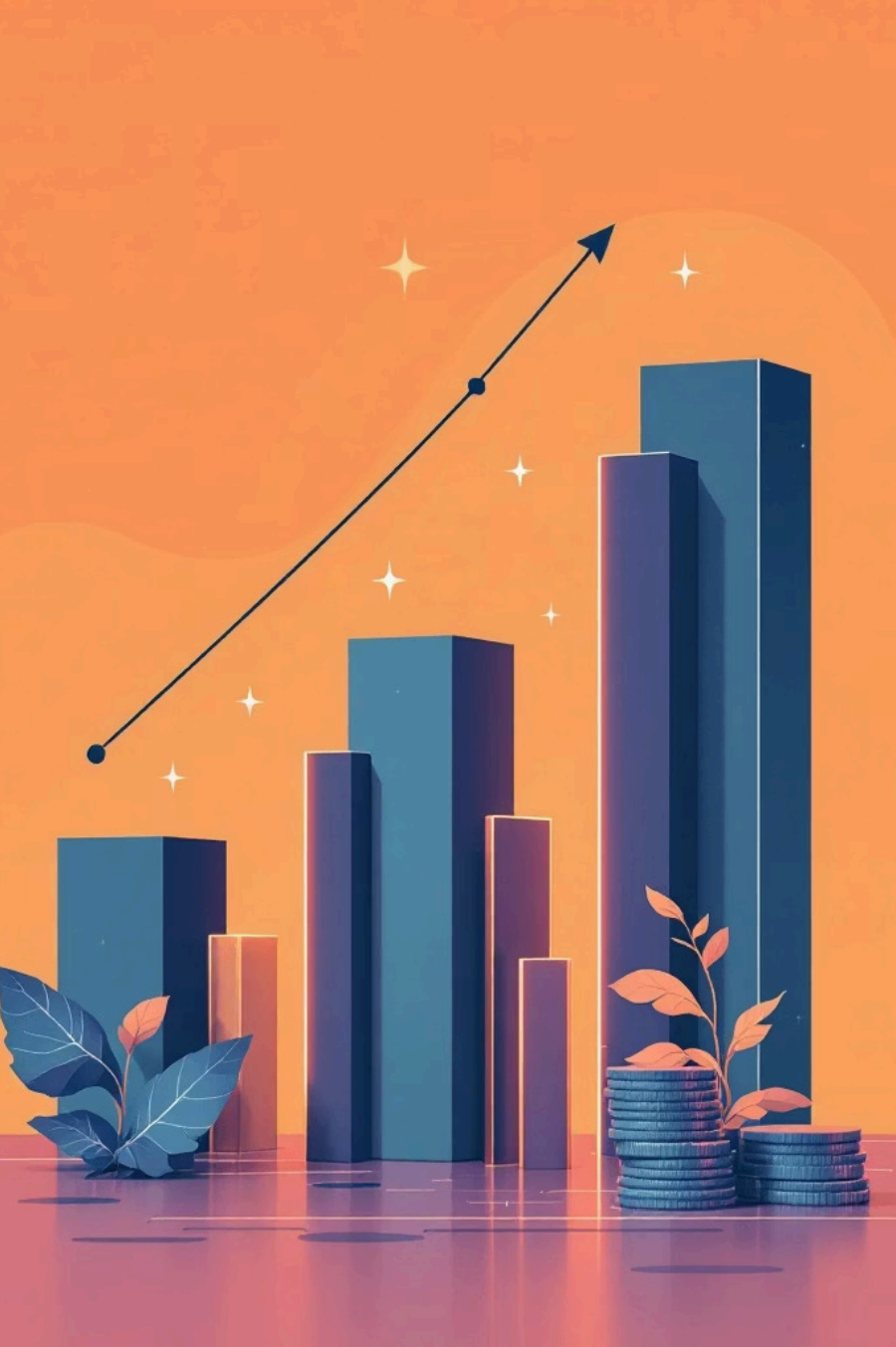
Feature Engineering

Created age groups and purchase frequency

05

Database Integration

Loaded to PostgreSQL for SQL analysis



SQL Analysis: Revenue Insights

Gender Revenue

- **Male:** \$157,890
- **Female:** \$75,191

Male customers generated 2.1x more revenue

Age Group Revenue

- **Young Adult:** \$62,143
- **Middle-aged:** \$59,197
- **Adult:** \$55,978
- **Senior:** \$55,763



Product Performance



Top Rated

Gloves (3.86 avg)

Sandals (3.84 avg)

Boots (3.82 avg)



Most Discounted

Hat (50%)

Sneakers (49.66%)

Coat (49.07%)



Best Sellers

Blouse (171 orders)

Jewelry (171 orders)

Pants (171 orders)



Shipping & Subscription Analysis

Shipping Comparison

- **Standard:** \$58.46 avg
- **Express:** \$60.48 avg

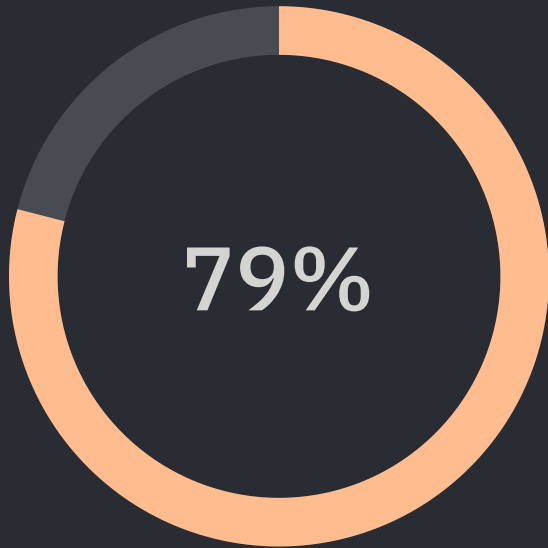
Express shipping customers spend 3.5% more

Subscription Impact

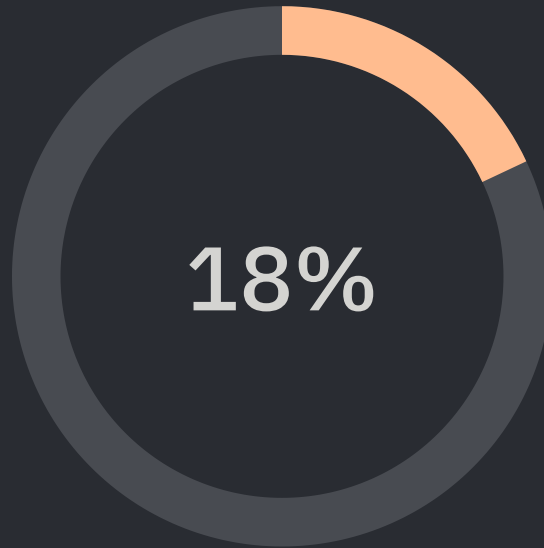
- **Subscribers:** 1,053 customers
- **Non-subscribers:** 2,847 customers

Non-subscribers generated 2.7x more total revenue

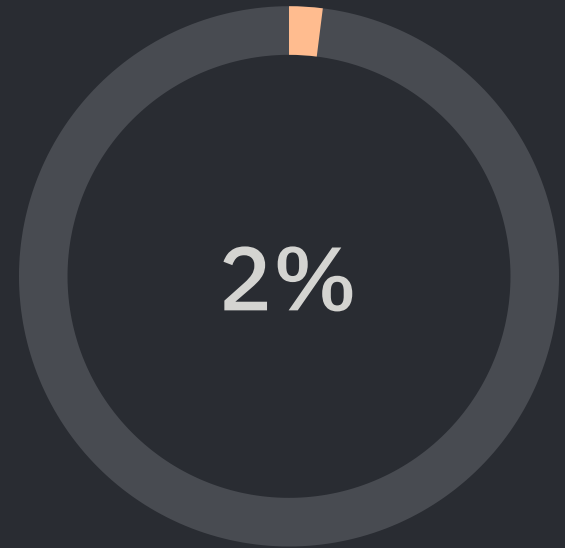
Customer Segmentation



Loyal
3,116 customers



Returning
701 customers



New
83 customers



Power BI Dashboard

KPIs Tracked

- 3,900 customers
- \$59.76 avg purchase
- 3.75 avg review rating

Visualizations

- Revenue by category
- Customer demographics
- Subscription status

Key Findings

- Male customers drive 2.1x revenue vs. female
- Young adults generate highest revenue (\$62K)
- Express shipping correlates with 3.5% higher spend
- 839 customers use discounts but spend above average
- 97% of customers are loyal or returning

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers



Customer Loyalty Programs

Reward repeat buyers to move into "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and express-shipping users